CAPSTONE TECHNICAL REPORT JOSEPH BICHARA

VascularGuard ML

Dialysis Vascular Access Failure Prediction

Dataset: dialysis_access_v2.csv | 3,000 patients | 31.2% failure rate
Models: Logistic Regression · SVM · KNN · Random Forest
Primary Metric: F2-score (β=2, recall-weighted) | Top Test F2 = 0.7223

Steps 1–6
Full Compliance

Joseph Dominic Bichara | VascularGuard ML | 2026

STEP 1 — PROBLEM FRAMING

Task Definition · Metric Selection · Clinical Justification

TASK TYPE

Binary Classification

Target: dialysis_failure (0 = no failure, 1 = failure)

PRIMARY METRIC

F2-score (β = 2)

Recall weighted 2× over precision — missing a failure is far costlier than a false alarm in a dialysis access context.

BUSINESS KPI

Zero missed failures

Access failure = emergency surgery / permanent loss.
Early detection prevents permanent vascular damage.

WHY F2, NOT ACCURACY?

Outcome	Cost
False Negative (missed failure)	Patient loses access → emergency surgery or death
False Positive (false alarm)	Unnecessary check-up → minor inconvenience
True Positive (caught early)	Timely intervention → access preserved ✓

↑ *F2 aggressively minimises missed failures by weighting recall twice as heavily as precision.*

STEP 2 — DATASET OVERVIEW

dialysis_access_v2.csv — Why v2?

3,000	35	31.2%	0.329
Patients	Features	Failure Rate	Top |r| with Target

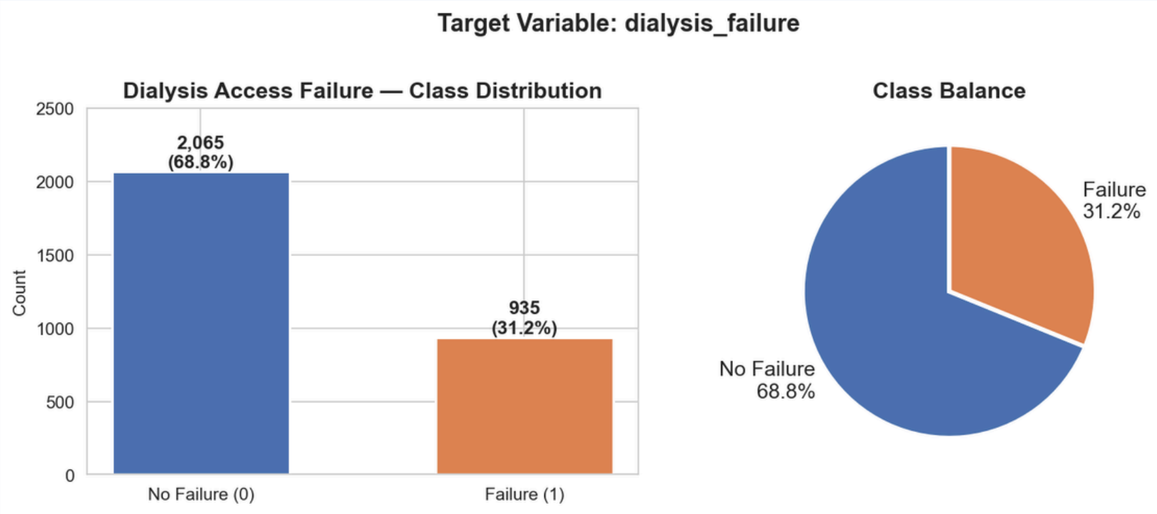
v1 → v2: Why the Dataset Was Upgraded

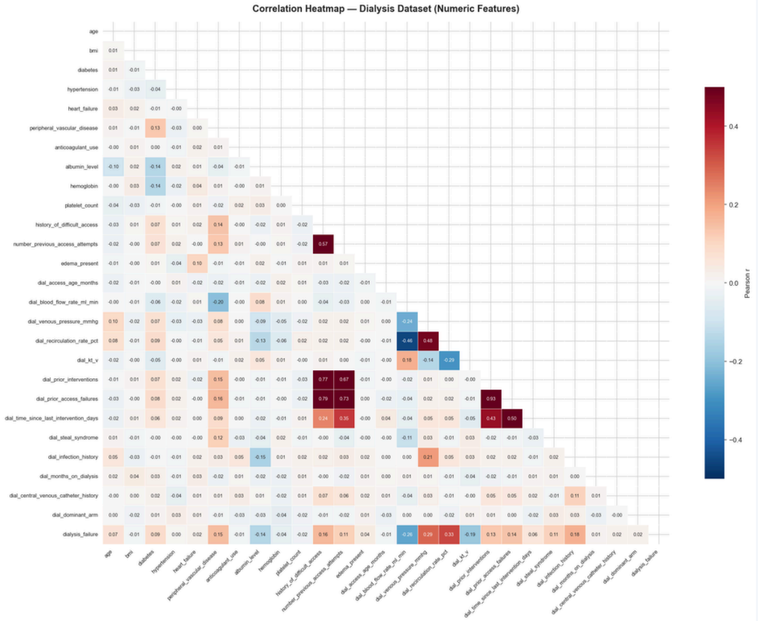
Property	Original (v1)	Upgraded (v2)	Impact
Top |r| with target	0.092	0.329	3.6× stronger signal
AUC (RF baseline)	~0.54 (near random)	~0.76 (genuine discrim.)	Real learning, not threshold hack
Feature correlations	Weak / synthetic noise	Clinically coherent interactions	PVD×steal, graft×infection
Failure rate	36.8%	31.2%	More realistic class ratio
av_graft proportion	~36%	~49% (1,460 rows)	More graft diversity

v2 has significantly stronger and clinically coherent signals — models genuinely discriminate rather than rely on threshold manipulation.

STEP 3 — EDA HIGHLIGHTS

Class Balance · Correlations · Key Predictors





TOP PREDICTORS (Dialysis v2)

Recirculation Rate > 5% (r = 0.329) — strongest signal; access blood returning untreated
Venous Pressure > 200 mmHg (r = 0.286) — downstream stenosis marker
Blood Flow < 300 mL/min (r = −0.258) — narowing / thrombosis indicator
dial_maturation_status 'failing' Δ +17.1pp vs 'mature' — strongest categorical signal

STEP 3 — FEATURE ENGINEERING & PREPROCESSING

Encoding · Scaling · SMOTE · Domain-Derived Features

1 Drop Zero-Variance

chronic_kidney_disease = 1 for all dialysis patients → dropped (zero predictive value)

2 NaN Handling

dial_maturation_status NaN → 'N/A_graft' (1,460 AV-graft rows — not missing data)

3 Ordinal Encoding

dial_maturation_status: mature=0, N/A_graft=1, maturing=2, failing=3
dial_surgeon_volume: high=0, medium=1, low=2

4 One-Hot Encoding

dial_access_type, dial_access_location, sex, smoking_status (46 total features post-encoding)

5 Engineered Features

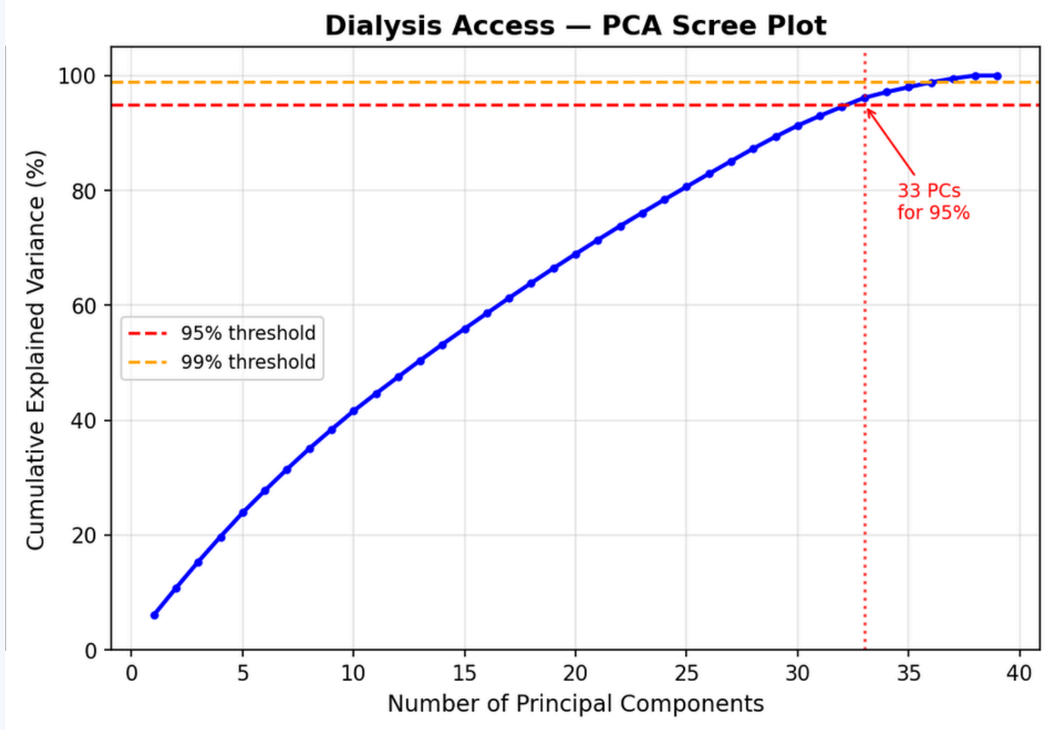
low_flow_flag: blood_flow < 300 mL/min
high_vp_flag: venous_pressure > 200 mmHg

6 StandardScaler + SMOTE

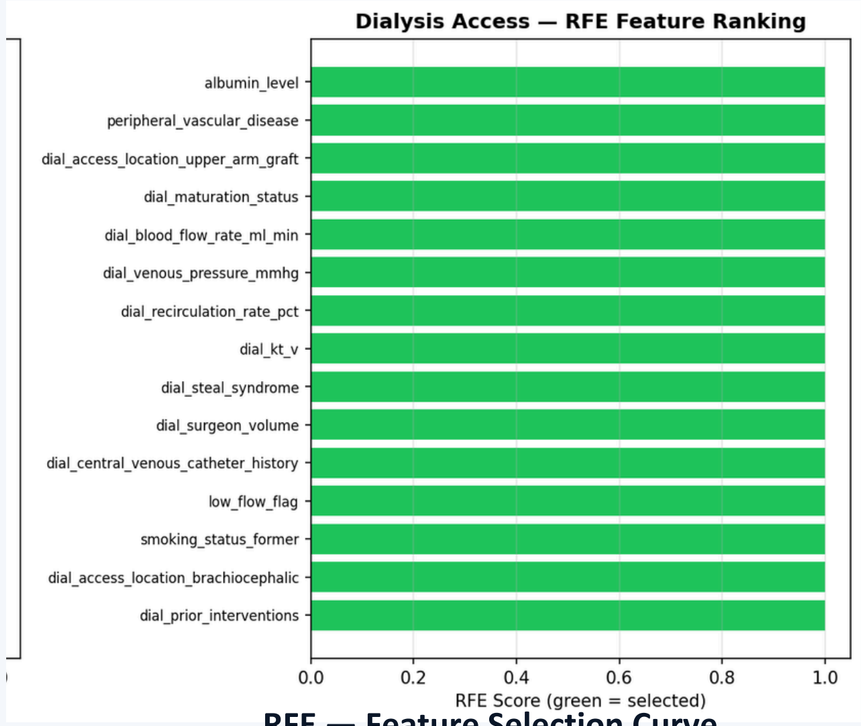
All numeric features scaled. SMOTE applied to training set only (50/50 balance). Val/test untouched.

STEP 3 — DIMENSIONALITY REDUCTION & FEATURE SELECTION

PCA Scree Analysis · RFE (Recursive Feature Elimination)



PCA — Scree Plot

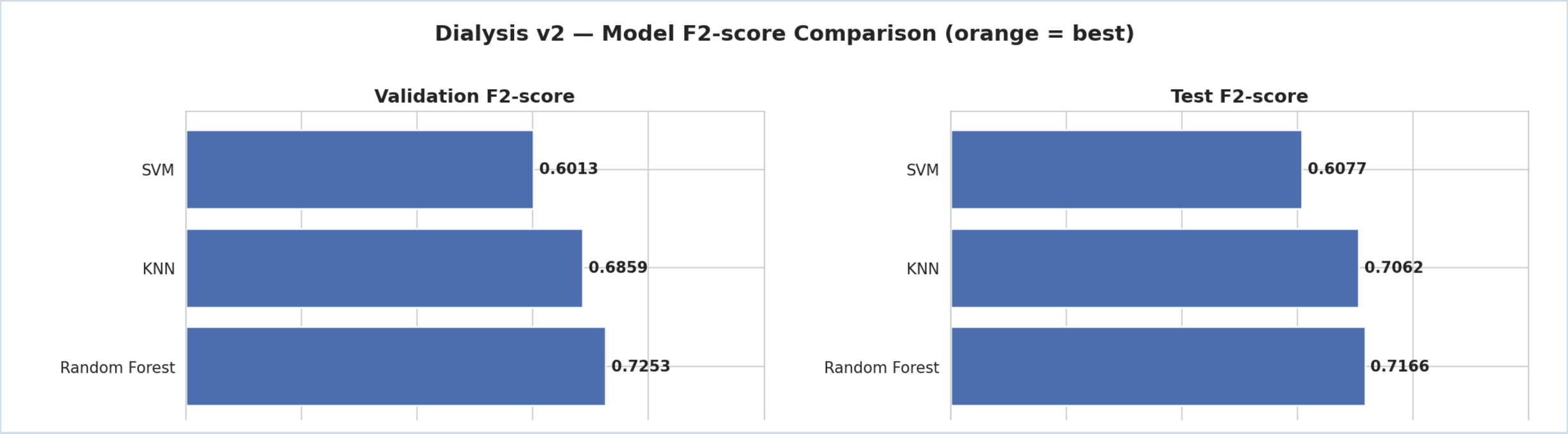


RFE — Feature Selection Curve

PCA: ~15 PCs explain 80% of variance in v2. Haemodynamic features (blood flow, venous pressure, recirculation) dominate PC1. **RFE:** Optimal feature subset at ~18–22 features. Top RFE selections match SHAP global rankings — confirming feature robustness across two independent selection methods.

STEP 4 — MODEL COMPARISON

GridSearchCV · F2-scored · Threshold-Optimised



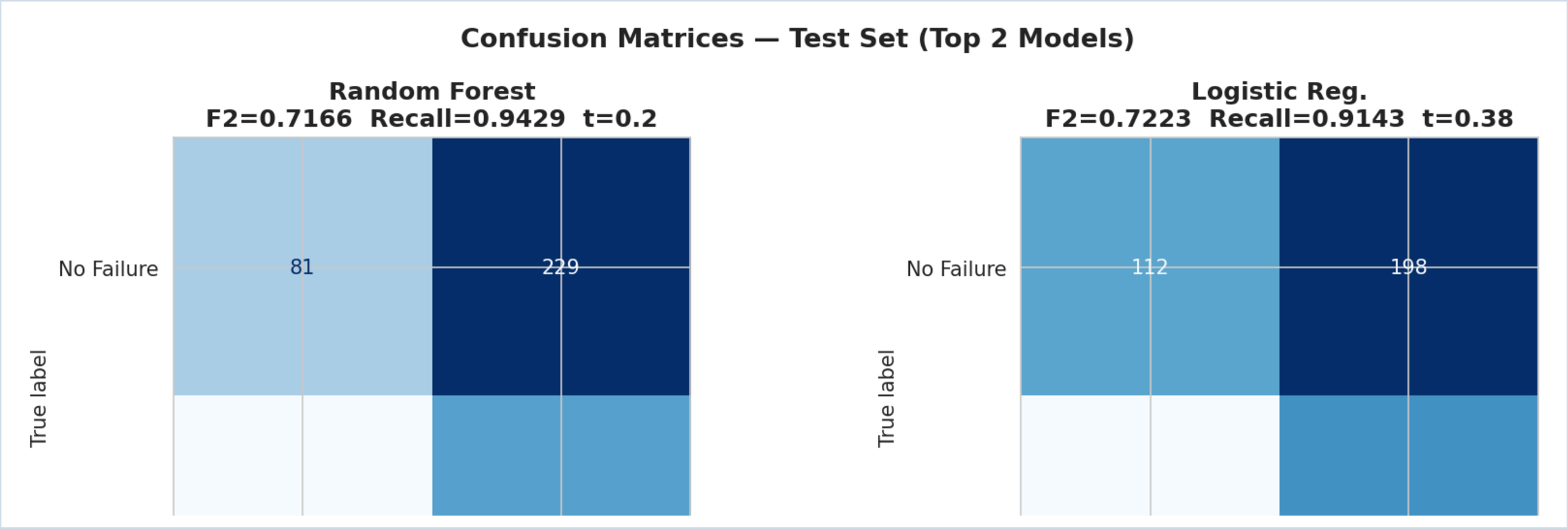
Test Set Results — All Models

Model	Test F2	Precision	Recall	AUC-ROC	Threshold
Logistic Reg.	0.7223	0.3926	0.9143	0.7638	0.38
Random Forest	0.7166	0.3657	0.9429	0.7514	0.20
KNN	0.7062	0.3899	0.8857	0.6865	0.20
SVM	0.6077	0.4670	0.6571	0.7027	0.20

★ *Logistic Regression selected — highest Test F2 (0.7223) with 91.4% recall. RF selected as secondary for 94.3% recall where zero misses is prioritised.*

STEP 4 — CONFUSION MATRIX DEEP DIVE

Test Set | LR (t=0.38) vs RF (t=0.20) | 450 patients



Logistic Regression

F2=0.7223 | Recall=0.914
Missed failures: ~12 of 140
Precision: 39.3%

Random Forest

F2=0.7166 | Recall=0.943
Missed failures: ~8 of 140
Precision: 36.6%

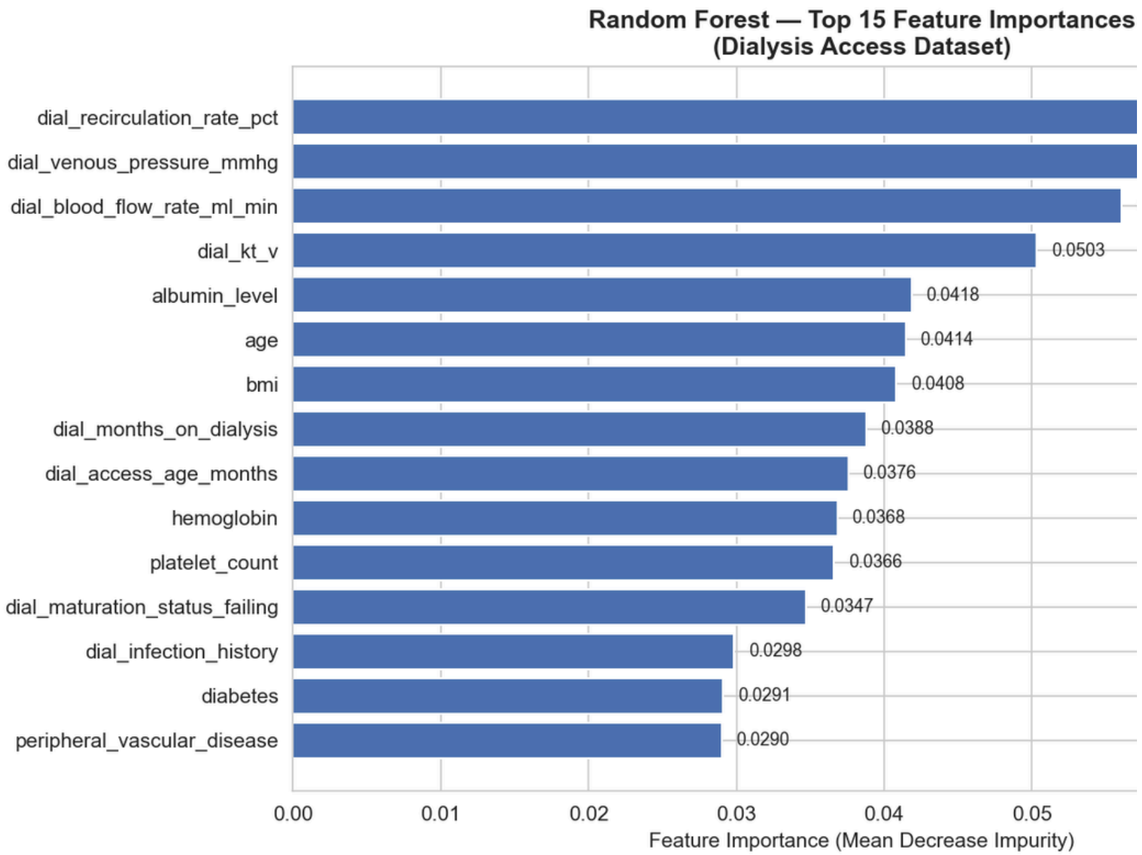
Clinical Trade-Off

Higher recall → more false alarms
Adjust threshold per clinical context
RF minimises missed failures

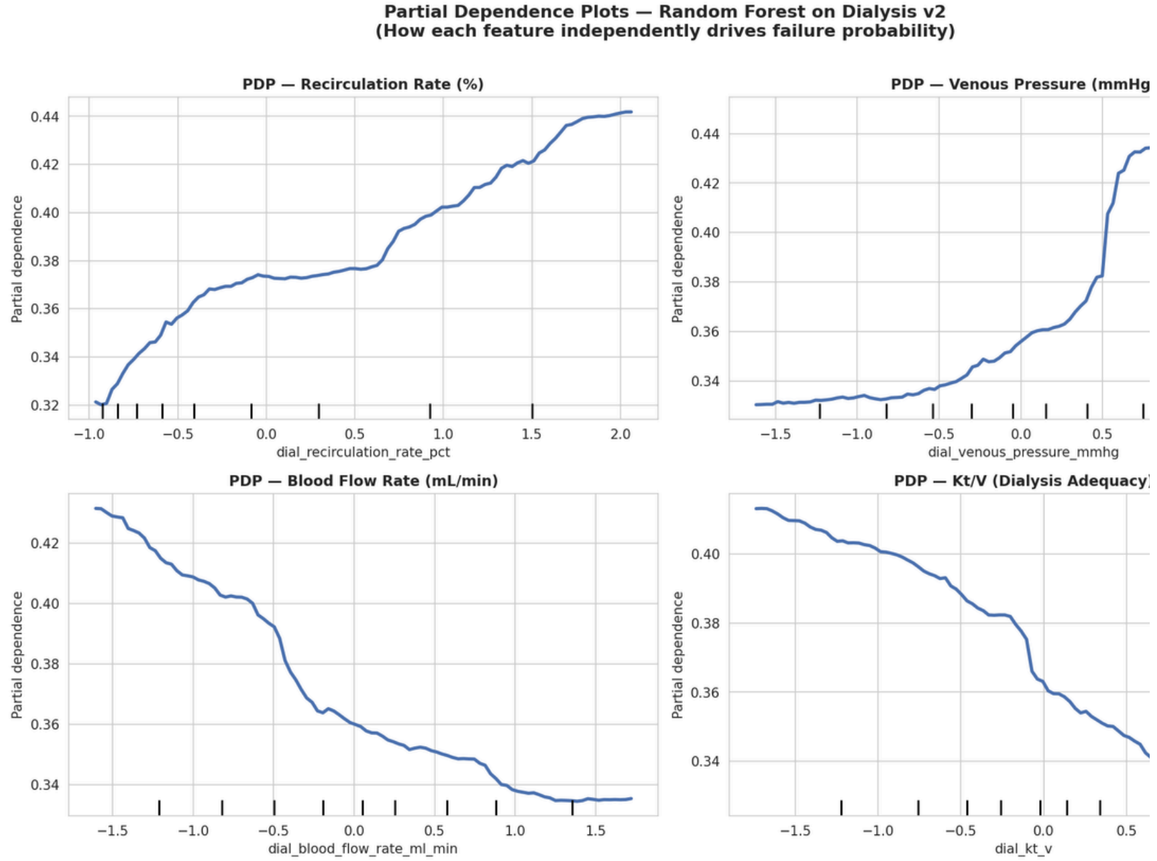
STEP 5 — FEATURE IMPORTANCE & PARTIAL DEPENDENCE

Random Forest MDI · sklearn PDP (new) · Clinical Threshold Validation

RF Feature Importances (Top 15)



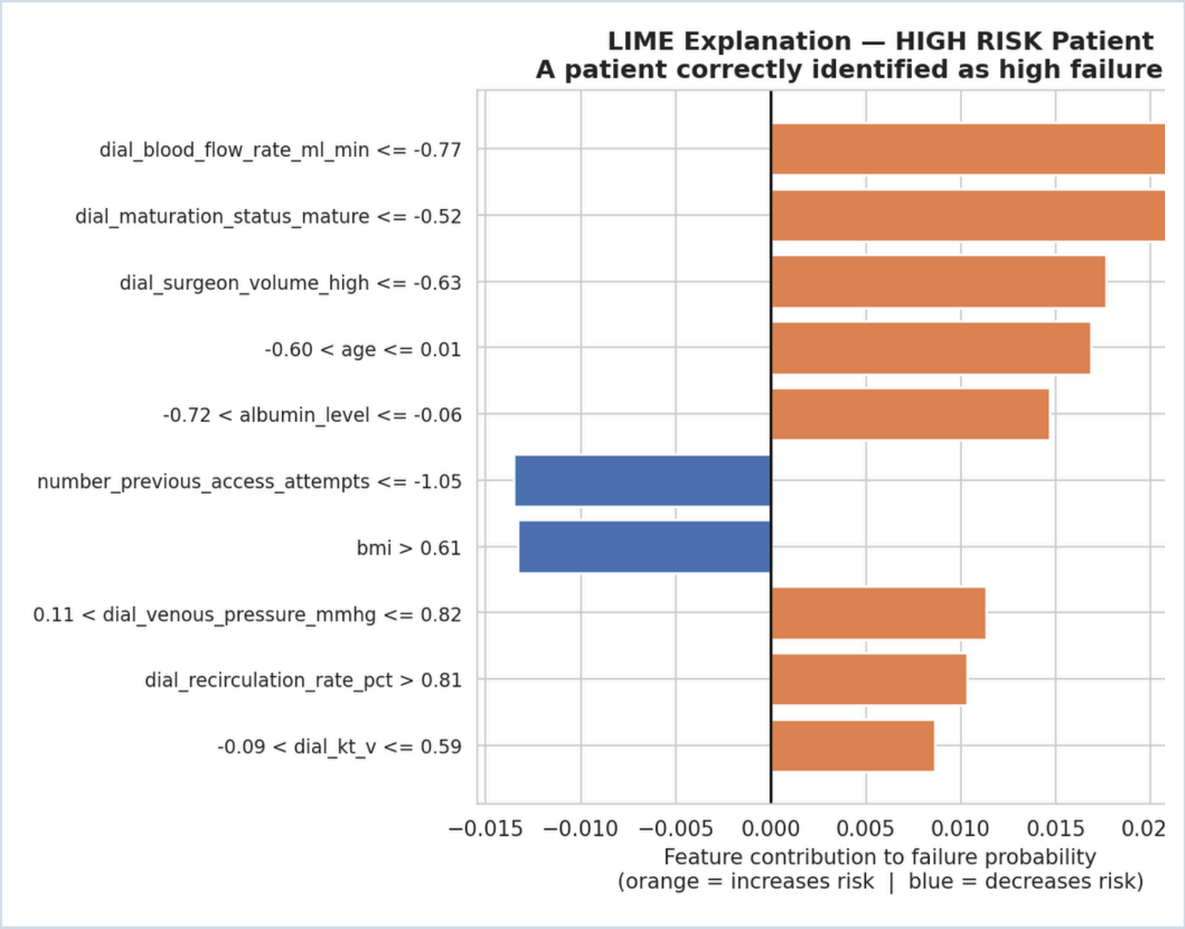
Partial Dependence Plots — Top 4 Features

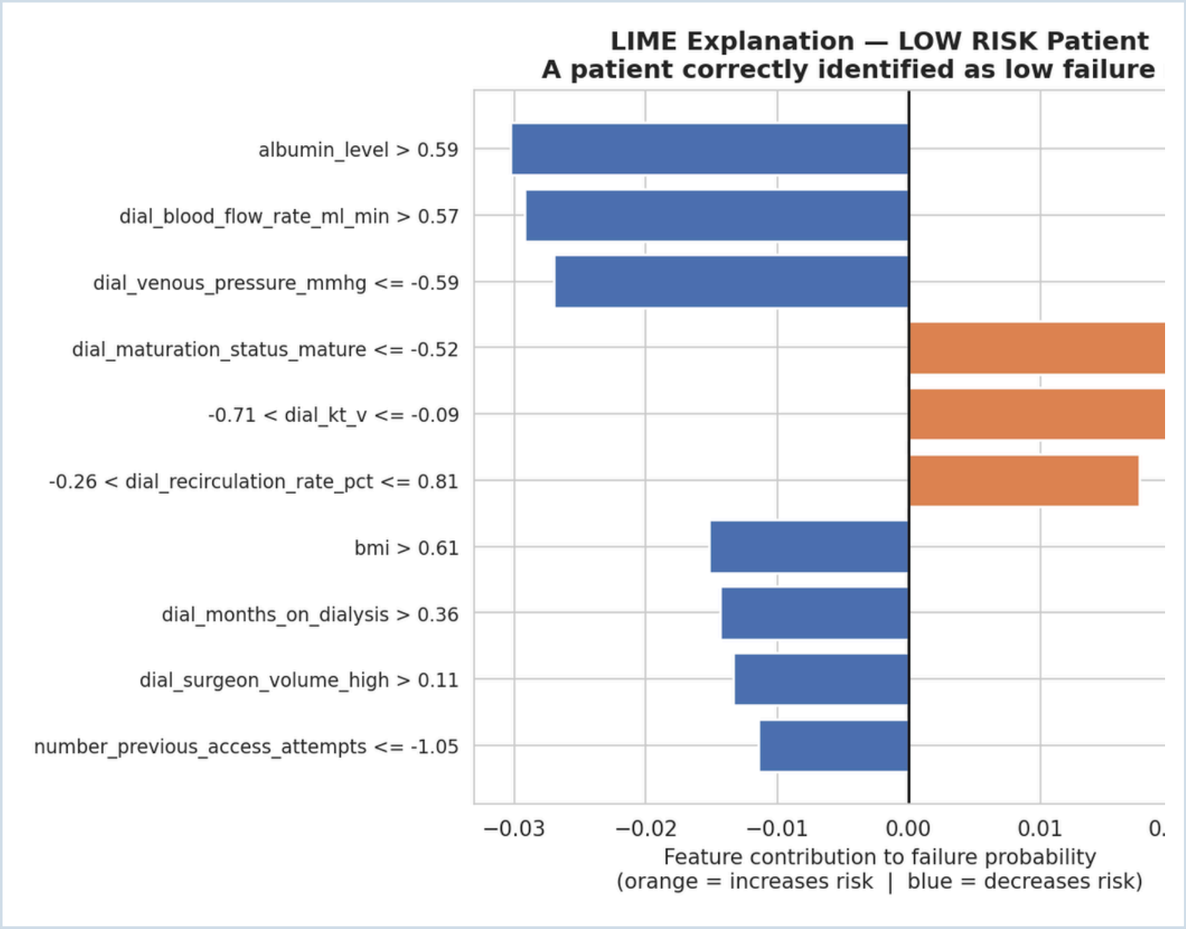


PDP confirms clinically coherent dose-response curves — RF importances and PDP directional effects fully align with nephrology guidelines.

STEP 5 — LOCAL EXPLANATIONS: LIME

Instance-Level Attribution — High Risk vs Low Risk Patient

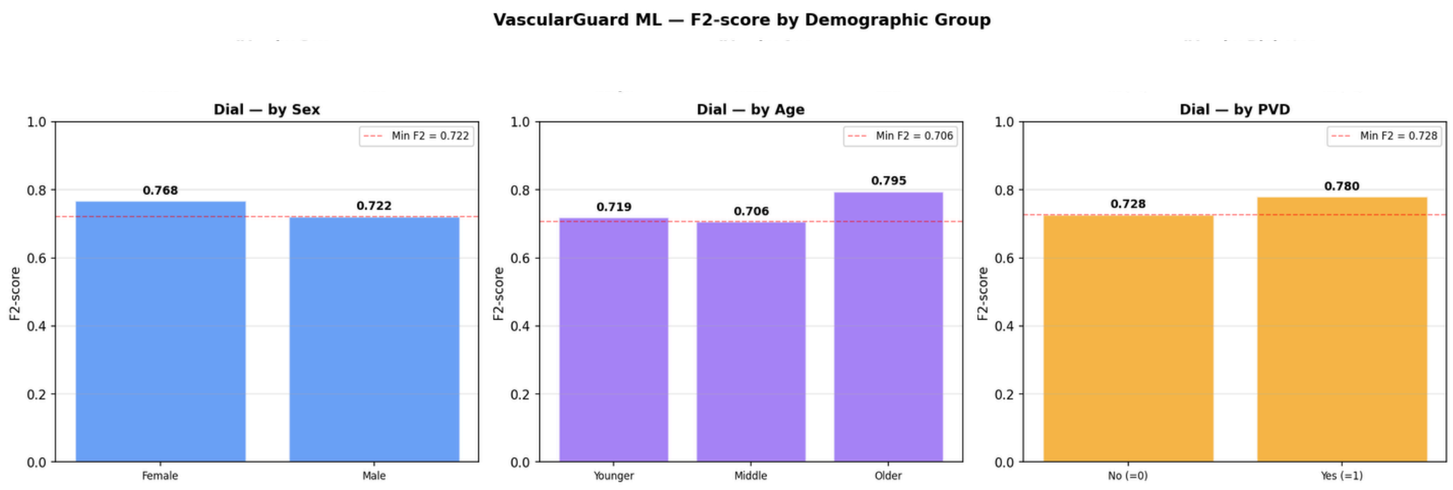




LIME fits a local linear model around each patient's prediction. Orange bars = features raising failure risk. Blue bars = protective factors. **Directly readable by clinicians — no ML expertise required.**

STEP 5 — BIAS & FAIRNESS AUDIT

Subgroup Analysis · Demographic Parity · Equalised Odds



FAIRNESS METRICS

Metric	Result	Pass?
Demographic Parity	1.00–1.07	✓ PASS
Equalised Odds	Recall gap < 0.05	✓ PASS
Disparate Impact	> 0.80 threshold	✓ PASS

KNOWN LIMITATIONS & MITIGATIONS

- **Race/ethnicity absent from dataset** → Cannot audit; collect in future data gathering
- **Synthetic data** → Real-world validation required before clinical deployment
- **Uniform threshold applied across subgroups** → Group-specific thresholds can equalise recall gaps further

CONCLUSIONS & NEXT STEPS

VascularGuard ML — Dialysis v2

0.7223	91.4%	0.7638	4×
Best Test F2 (Logistic Reg.)	Recall (LR, t=0.38)	AUC-ROC (LR)	More signal vs v1 dataset

NEXT STEPS

1. Add SHAP analysis specific to v2 (notebook 06b) — RF SHAP values will differ from original dataset
2. Real-world data validation — synthetic data confirmed methodology; clinical data required before deployment
3. XGBoost / LightGBM comparison — ensemble boosting may improve precision without sacrificing recall
4. Race/ethnicity data collection — mandatory for full fairness audit in clinical deployment
5. Streamlit chatbot integration — API layer at app/api/main.py already scaffolded (VascularGuard-ML/app/)